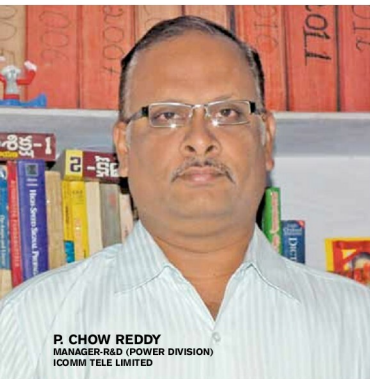


“I am very keen on using Raspberry Pi for military applications”



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What are the most exciting trends in R&D?

There is immense requirement of USB, Internet connectivity and RS232 in power distribution boards (PDBs) and automatic changeover against mains failure (AMF) panels for their access to the media and system connectivity. Apart from this, I have identified the utility of ruggedised, dedicated computing systems for military applications. For all military applications, one single solution would be Raspberry Pi.

What makes Raspberry Pi work for such serious applications as in military and defence?

We have been working on Raspberry Pi. It consumes less power, and is virus-free (since it is Linux-operated) and rugged (as it uses solid-state memory in the form of a SD card compared to a hard disk). It is small and minimally configured, and can fit into a small cabinet or on the back of a monitor. Its HDMI is faithful in giving a marvellous picture quality. Above all, it is very inexpensive and customers accept it.

What kinds of Raspberry Pi projects could help design engineers?

We have raised proposals of the projects based on Raspberry Pi. In parallel, we have also developed a small applica-

tion that involves data storage memory, display, RS232 and input/output (I/O) port handling. All this is operated on Python and has been found to be operating up to the mark. One thing that an engineer has to keep in mind is that it lacks real-time clock (RTC).

Personally, I am very keen on using Raspberry Pi for military applications. There are numerous other applications and fields where it can be utilised. It has a big future ahead and if its cost reduces further to the declared figure of \$25,000, one can think of replacing microcontrollers with Raspberry Pi.

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Could you elaborate on incorporating USB, Internet connectivity or even RS232 in PDB and AMF panels by utilising a Raspberry Pi?

AMF is used for changeover from mains to a diesel generator (DG) set and vice versa depending on the conditions. In these panels, USB is required to store the logged data like power fluctuations, DG run time and fuel filling updates to estimate the consumption at all load levels. All this is saved into a pen drive.

When it comes to RS232, I personally would require it for communication between other panels, LCD instrumentation displays and annunciation through GSM/GPS. Internet connectivity is used to take the stored data for a view on the performance of the panel, loads and power sources. All this exists as hardware in Raspberry Pi. Additionally, all the logical decisions, and activation of relays, contactors and solenoids are possible through software.

Could you tell us about a recent cost-effective solution developed by your team, deviating from existing approaches?

Mil-grade air-conditioners are designed and manufactured in monopoly by very few companies. These are costly